BMET3997/9997
Physiological Signal Sensing and Processing
Major project

THE UNIVERSITY OF SYDNEY

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We acknowledge the tradition of
custodianship and law of the Country on which
the University of Sydney campuses stand.
We pay our respects to those who have cared
and continue to care for Country.

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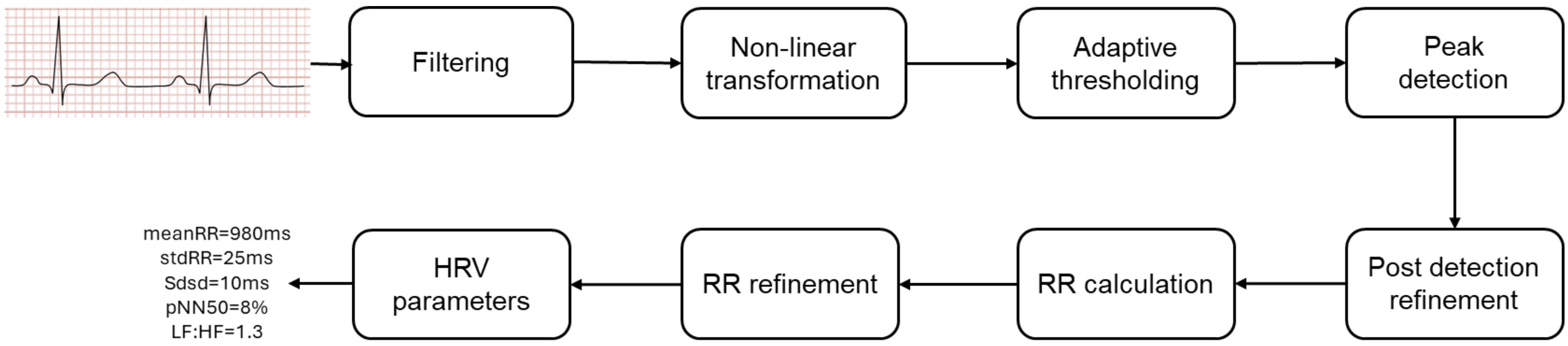
Major project

- **You will work in groups of 3 students.**
  - Currently 70 students enrolled so 23 groups.

- **The biomedical challenge is to develop a system that automatically identifies QRS complexes and estimates heart rate variability parameters from single lead electrocardiographic (ECG) signals.**

Major project – a possible approach



Major project

- **Each group will need to achieve the following**
  - Use provided *training data*, develop a QRS detection algorithm and HRV calculation algorithms that will process the provided training and test data. The format of test data is the same as the training data
  - Submit your QRS detection times and your estimated HRV parameters for the *test data*. We will then assess scores against the test data and calculate the performance metrics and inform you of the performance.
  - Each group will get a maximum of 5 submissions sent to the lab. tutor between weeks 7 and week 13.
  - The best set of performance metrics for each group for the test data will be displayed on Canvas and each group will compete to get the highest score.

Major project

- **Each group will need to achieve the following**
  - Each group will submit the following
    - A powerpoint presentation describing your approach and best solution to date (Week 13, lecture/lab time)

- **Each individual will need to achieve the following**
    - A report summarising your group approach and your individual contributions (Tuesday, 1st exam period)
      - Including Python/Matlab code implementing your group's final solution. This code must be in a format that I can run, as I will check that it produces the scores you submit

Major project – training data

- **Available on Canvas under the Major Project Module** ***ProjectTrainData.mat***

- **Available on Canvas under the Major Project Module ProjectTrainData.mat. It contains two cell arrays**
  - **ECG**: a cell array with 35 overnight ECG each sampled at 100Hz and signal amplitude in uV. ECG{1} contains the ECG signal for overnight recording 1, ECG{2} contains the ECG signal for overnight recording 2, etc. Note that each night has a different length.
  - **QRSexpert**: a cell array with the expert validated QRS points for each of the ECG recordings. QRSexpert{1} contains the QRS detection points (in samples) for overnight recording 1. QRSexpert {2} contains the QRS detection points for overnight recording 2, etc. Note that each night has a different length.

Major project – testing data

– **Available on Canvas under the Major Project Module** ***ProjectTestData.mat, ProjectTestDataAnalysis.mat***

– **The *ProjectTestData.mat* file contains 35 overnight ECG (different to training data!).**

– **The *ProjectTestDataAnalysis.mat* contains a cell array for the QRS detections and arrays for the HRV parameters.**

Major project – *ProjectTestDataAnalysis.mat*

- **QRS**
  - **As they currently are**
    - QRS{1}'=[100,200,300]
  - **Once ready for a submission, you will need to change the variables to something like**
    - QRS{1}'= [56, 148, 276, 512,…..,]
  - **I expect there will be between 20-50k detections for each of the 35 recordings.**

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- **HRV parameters**
  - There is one array for each HRV parameter in ProjectTestDataAnalysis.mat and they are currently set to NaN. You will replace these values with your averaged values from above. For example, for record 1
  - As they currently are
    - avgRR[1]=nan; sdRR[1]=nan; RMSSD[1]=nan; pNN50[1]=nan; LF[1]=nan; HF[1]=nan; LF_HFratio[1]=nan;
  - Once ready for a submission, you will need to change the variables to something like
    - avgRR[1]=950; sdRR[1]=45; RMSSD[1]=34; pNN50[1]=12; LF[1]=605; HF[1]=495; LF_HFratio[1]=1.2;

And save it as new .mat file. Please DO NOT include the *ECG* variable from *TestingData.mat* in your submission. The new file name will contain your group number and submission number e.g. *ProjectTestDataAnalysisGroup2Submission3.mat*

We will use it to calculate a) sensitivity, positive predictivity and F1-score of your QRS detection points and b) mean absolute percentage error of your HRV parameters using the expert-based QRS detection points (which we will keep secret) and report back to you.

Towards end of the semester, I may also get you to submit your training set QRS detections and HRV parameters of your best system so we can assess the training set performance of your system

Major project – Key points

You are free to use any of the toolboxes in Matlab and libraries in Python except for toolboxes or libraries that relate to QRS detection or HRV calculation. If you are unsure ask Siying of myself.

Python implementations must be able to process the *.mat file as input and produce the required *.mat as output.

- We will provide code that will help with this conversion

Major project - submissions

Each group will get a maximum of 5 submissions sent to the lab. tutor between weeks 7 and Friday 5pm week 13.

If your submission is in the wrong format, then we will let you know and you can submit again without using up one of your submissions.

- **It is up to your group to make sure you have the correct format!**

Major project – Key points

Major areas of technical work to be finished by week 13 include

- **Implement a QRS detector as a first stage**
- **Implement a second stage algorithm that assesses QRS detection points from the first stage and attempts to correct possible false positives and false negatives (post-detection refinement).**
- **Implement algorithms for calculation of required HRV parameters in the presence of imperfect or missed QRS detections.**

Non technical work

- **Final report (Tuesday 1st week exam period)**
- **Presentation (Monday week 13)**

Formats + rubics for the report and presentation will be provided later in the semester

Division of duties

- **It is up to your group to decide who does what. Here is one possible division of duties**
  - One person working on QRS detector first pass
  - One person working on QRS post detection refinement- trying to find QRS detections that are potentially missed by first pass
  - One person working on HRV calculation

Notes for BMET9997 students

- **Your QRS detector must process causal data**
  - i.e you can only use ECG data from past (not the future)
  - BMET3997 students can use ECG data from the past and the future

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